

Attractor Representations for Manifolds in Neural Circuits

For Application to the SWC-GCNU PhD Programme

Tim Lin

University of Cambridge
Supervisor: Prof. Guillaume Hennequin

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Agenda

Motivation

Controller Setup

CAN Theory: What is it and what have we learned?

Ring Attractor Theory

Attractor Network as Optimized Solutions

My Dissertation Project

Training Attractors with minimal assumptions

Learning motor control with an attractor

Motivation

The brain is thought to hold internal representations of variables, in order to solve the below problem:

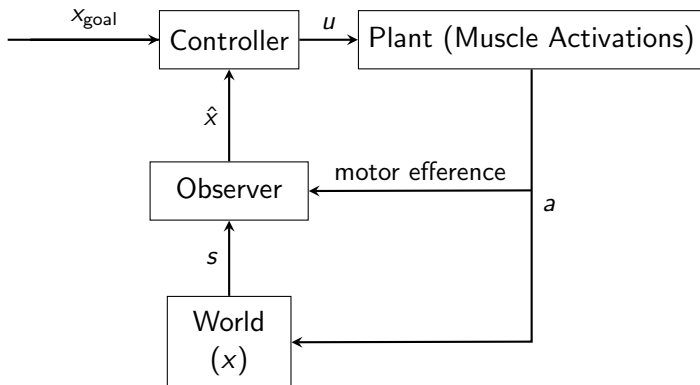
Given x_t , C , find $a_t^* \sim f(x_t, C)$

such that $\sum_{\tau=t} R(x_\tau, a_\tau^*, C)$ is maximised.

where x_t is a current state, C is the context (which may contain x_{goal}), and a_t^* is the action. R is the reward function, which can be additionally dependent on $\{\gamma_\tau\}_\tau$.

Motivation

In a classical control perspective, we only observe low quality s from the world and hence must have an internal state estimate.



Motivation

Now, we must notice that x_t can live in different spaces!

- ▶ Spatial cognition: 2D space, *grid cells*, *place cells*
- ▶ Head orientation: 1D ring, *HD system in mice and fly*
- ▶ Head *rotation*: $SO(3)$, *bat presubiculum*, *mice SC (?)*
- ▶ Object manipulation: CGA, *motor hierarchy (?)*

... and various graph structures related by discrete actions (see Whittington et al. 2022 Nature).

When the state space is continuous and locally Euclidean, they are *manifolds* and we denote $x_t \in \mathcal{M}$.

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CAN Problem Setup

We use a set of neural states $g_t \in \mathbb{R}^d$ to *represent* x_t . Described in general terms,

$$\dot{g} = F(g, u, t) \quad (1)$$

where $u \in \mathbb{R}^{d_M}$ lives in the tangent space of the manifold.

This continuous time system is often approximated by the following discrete time system, and we assume that the system is invariant:

$$g_{t+1} = \mathcal{F}(g_t, u_t) \quad (2)$$

Ring Attractor Theory (Zhang, 1996)

Similar to the classical setup, investigate the following system (e.g. Amari, 1972)

$$\begin{aligned}g &= \sigma(v) \\ \tau \dot{v} &= -v + W(t) * g\end{aligned}$$

where $v, g \in \mathbb{R}^d$ are the neural activity (e.g. voltage) and the neural state after nonlinearity. $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$

$W(t) \in \mathbb{R}^{d \times d}$ gives the weight distribution to be convolved with the neural state.

How was the ring attractor set up?

Parametrize neural response as $g(\theta, t)$ ¹

Induce rotation invariance by a circulant weight kernel $W(\theta, \theta', t)$ that only depends on the difference $\theta - \theta'$.

$$W * g(\theta, t) \rightarrow \int_{\Theta} W(\theta - \theta', t) g(\theta', t) d\theta'$$

Then the weight distribution is a function defined in the angular domain, $W(\theta, t) = W_{\text{odd}}(\theta, t) + W_{\text{even}}(\theta, t)$

When $W_{\text{odd}} = 0$, the system can settle to some static attractor state with appropriate σ (minimum energy in the Lyapunov form).

¹Consider infinite neural coding resolution

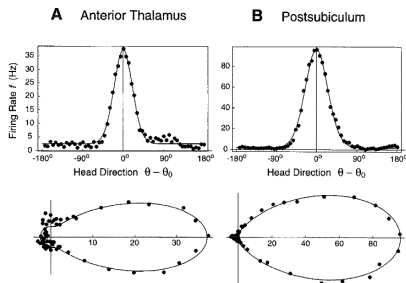
Attractor network as a solution

Candidate directional tuning curve: ²

$$g(\theta) = A + B \exp(K \cos(\theta - \theta_0))$$

Candidate nonlinearity: ³

$$\sigma(v) = a \ln^\beta(1 + \exp(b(v + c)))$$



²Circular normal distribution, parameters fitted from observed neural firing

³Exponential decay behavior in the negative half and power function in the positive half

Solve for weight distribution in the Fourier domain:

$$\dot{v}(\theta) = -v(\theta) + \int_{\Theta} W_{\text{even}}(\theta - \theta') g(\theta', t) d\theta' = 0$$

$$v(\theta) = W * g(\theta)$$

$$\hat{v}_n = \hat{W}_n \hat{g}_n \quad \forall n \in (-\infty, \infty)$$

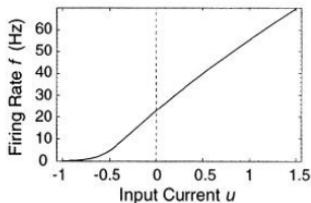
Adding a small weight regularization to encourage finite-energy solution ($\|W\|_2^2 < \infty$), we can solve for the Fourier coefficients of the weight distribution.

$$\hat{w} = \arg \min_{\hat{w}} \sum_n |\hat{v}_n - \hat{w}_n \hat{g}_n|_2^2 + \lambda |\hat{w}_n|_2^2$$

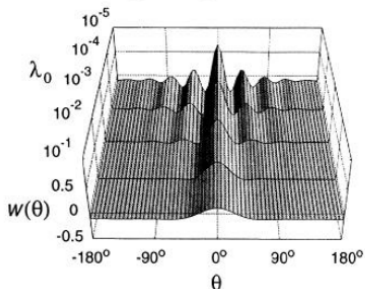
$$\hat{w}_n = \frac{\hat{v}_n \hat{g}_n}{|\hat{g}_n|_2^2 + \lambda}$$

Optimized Weight Distribution

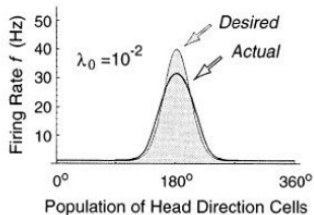
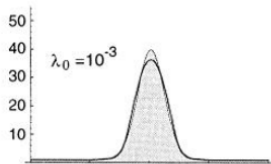
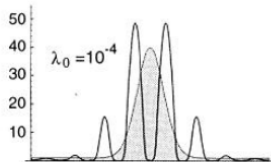
A Sigmoid



B Weight Regularization



C Stable States



Shifting mechanism

To achieve rigid bump shape when shifting along the manifold, we must ensure that the odd component of the weight distribution is its derivative ⁴.

$$W(\theta, t) = W_{\text{even}}(\theta) + \gamma(t) \frac{dW_{\text{even}}}{d\theta}$$

The relationship between $\gamma(t)$ and angular velocity $\omega(t)$ is given by the following equation ⁵:

$$\gamma(t)/\tau = \omega(t)$$

⁴This can be shown by dissecting a symmetric travelling profile

⁵This is to relate the speed at which the rigid profile is rotated to the

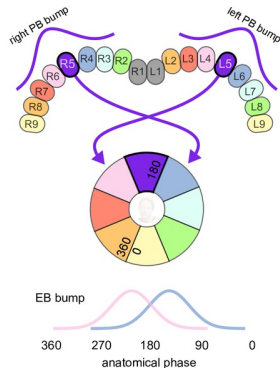
Biological implementation of the multiplicative gate

$W(\theta, t)$ describes the total weight distribution, which can be realized by two groups of cells:

Rep : Component of the cell population with symmetric connections, implementing $W_{\text{even}}(\theta)$

Input : Component of the cell population with asymmetric connections, effectively implementing $\frac{dW_{\text{even}}}{d\theta}$, dual tuned to the velocity signal.

This is called the copy-and-shift mechanism, and is almost directly implementable in the fly brain.



^aHulse et al., 2021

Attractor optimized for cells in the EC

In the above formulation, cells are defined by tuned angle.

When considering other systems (e.g., mammalian), it becomes less and less easy to do this type of definition.

EC = {grid cells, border cells, band-like cells, etc.}

Cueva and Wei (2018): Optimize directly for path-integration.

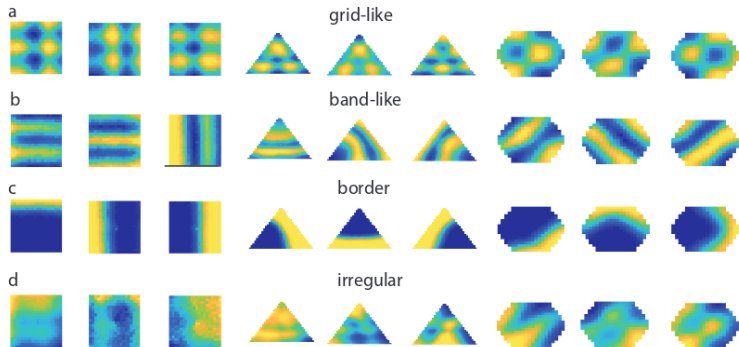
$$\tau \frac{dv_i(t)}{dt} = -v_i(t) + \sum_j W_{ij} g_j(t) + \sum_k^{N_{\text{input}}} W_{ik}^{\text{input}} u_k(t) + b_i + \epsilon_i(t)$$

$$y_j(t) = \sum_i^N W_{ji}^{\text{output}} g_i(t)$$

for loss function

$$\mathcal{L} = \langle \|y - y_{\text{target}}\|^2 \rangle_{T, N_{\text{out}} B} + \lambda_1 \langle \|W\|_2^2 \rangle + \lambda_2 \langle \|g\|_2^2 \rangle$$

Training Results



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CAN Problem Setup

For a neural dynamical system to be a CAN for a manifold, it should satisfy the following two postulates.

$$\text{Stability: } \mathcal{F}(g_t, 0) = g_t \quad (3)$$

$$\begin{aligned} \text{Path Integration: } \int_t^{T_1} u_\tau d\tau = \int_t^{T_2} \mu_\tau d\tau &\iff \\ g_{T_1} \mid \{u_\tau\}_{\tau \in [t, T_1]}, g_t = g_{T_2} \mid \{\mu_\tau\}_{\tau \in [t, T_2]}, g_t \end{aligned} \quad (4)$$

where integration is performed on the manifold, such as by consulting its Lie algebra.

Data and Training

Trajectories of actions are sampled by an AR(1) process

$$u_{t+1} = \alpha u_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma_{\text{AR}})$$

With this base dataset of trajectories, each gradient step is expressed as following.

$$\Phi_{e+1} = \mathcal{D}(\Phi_e, \mathbf{A}_e, \mathbf{P}_e)$$

where $\mathbf{A} = \{u_{1:T}^{(i)}\}_{i \in 1:B}$ and $\mathbf{P} = \{\mu_{1:T}^{(i)}\}_{i \in 1:B}$, such that

$$\left\| \int_1^T u_\tau^{(i)} d\tau - \int_1^T \mu_\tau^{(i)} d\tau \right\| < \sigma_x \quad \forall i$$

Loss Function

Similarity:

$$\left\| \int_t^T u_\tau d\tau - \int_t^T v_\tau d\tau \right\| < \sigma_x \implies \mathcal{F}(g_t, u_{t:t+T}) \approx \mathcal{F}(g_t, v_{t:t+T})$$

Contrast:

$$\int_t^T u_\tau d\tau \neq \int_t^T v_\tau d\tau \implies \mathcal{F}(g_t, u_{t:t+T}) \neq \mathcal{F}(g_t, v_{t:t+T})$$

Total loss:

$$\mathcal{L}_{\text{contrastive}} = \frac{\lambda}{\beta} \sum_{i=1}^B \phi(g_{t^*}^{\mathbf{A},(i)}, g_{t^*}^{\mathbf{P},(i)}) - \log \sum_{k=1}^B \exp\left(\frac{\phi(g_{t^*}^{\mathbf{A},(i)}, g_{t^*}^{\mathbf{P},(k)})}{\beta}\right)$$

Architecture

Segment the neural net into Rep and Input components, as in the Zhang model.

$$\begin{aligned} g_{t+1} &= \begin{bmatrix} g_{t+1}^{\text{rep}} & g_{t+1}^{\text{in},1} & \dots & g_{t+1}^{\text{in},J} \end{bmatrix}^{\top}, \quad \text{where} \quad (5) \\ g_{t+1}^{\text{rep}} &= \sigma(W_{rr}g_t^{\text{rep}} + \sum_{j \in [1,J]} u_t^j W_{ir}^j g_t^{\text{in},j} + b_r) \\ g_{t+1}^{\text{in},j} &= \sigma(W_{ri}^j g_t^{\text{rep}} + \sum_{j \in [1,J]} u_t^j W_{ij}^j g_t^{\text{in},j} + b_i^j) \end{aligned}$$

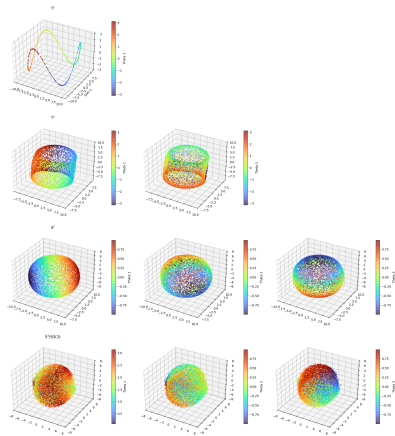
A more flexible setup can be used for a similar path integration problem, allowing for more complex modulation by actions.

$$W(u_t) \triangleq \text{MLP}(u_t)$$

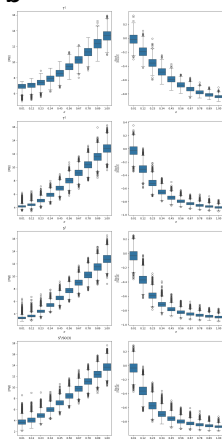
Results

Check structure and noise robustness of the learned attractor.

a



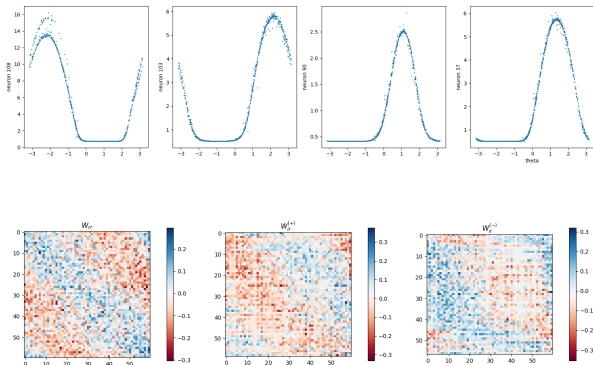
b



Comparison with Zhang's model

Additional constraints:

- ▶ Weight regularization
- ▶ Biologically plausible implementation (an input channel for each sign)



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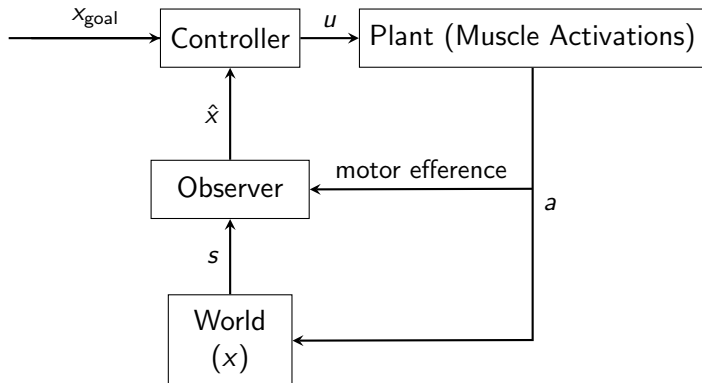
My Dissertation Project

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Revisit the Controller

This is just the setup from the first slide.



Using the attractor for motor control

Use our attractor as the observer here. Schematic on the next slide.

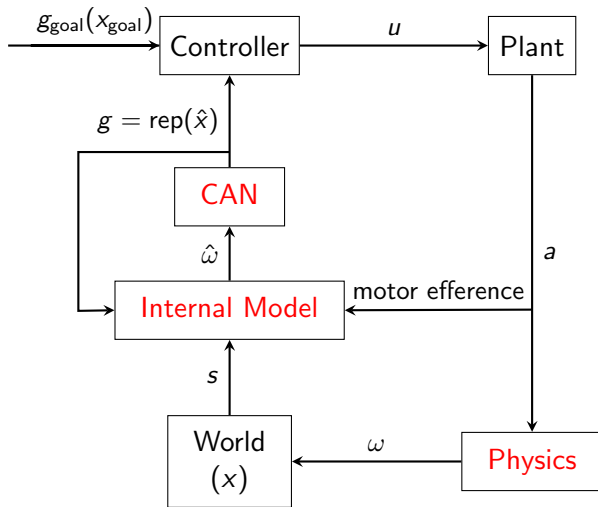
Notice that in this case the CAN is a purified dynamics model, which integrates inferred actions on the manifold. Note that $\hat{\omega}$ need not to equal ω , but it should act on the manifold consistently as ω acts on the world. The internal model is a lumped-together network which (1) resolve any misalignment between the estimated state and the true state implied by sensory feedback, and (2) transform motor efference into manifold actions.

Introduce two control settings: egocentric and allocentric.

allo: State representation is in the world frame. It should represent the current state and the goal represents the target state.

ego: State representation is in the self frame. It should represent a target transformation. The goal represents the null transformation, signifying no residual transformation.

Using the attractor for motor control

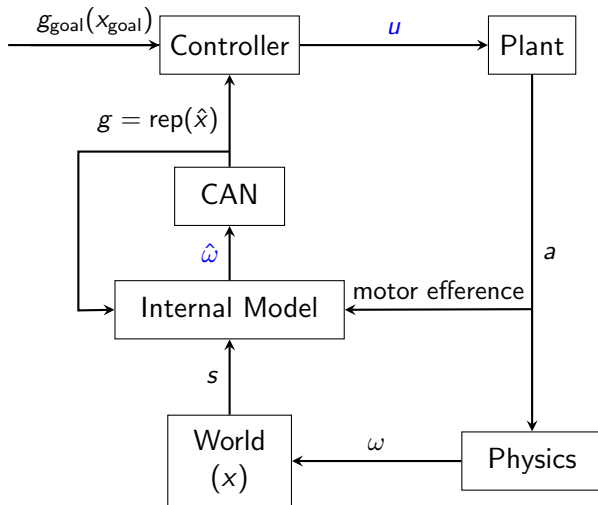


Training the attractor for motor control

For training these things, we consider only learning the controller and the internal model, and assume that a manifold-CAN is recruited properly for the structure of the behavioral problem.

One (naive) idea is to just train the system to settle the mismatch between the estimated state and the goal state, as well as resolve any residual actions (so to minimize $\hat{w}$ and u , as well as potentially $SE(g_{\text{goal}}, g)$)

Training the attractor for motor control



Training the attractor for motor control 2

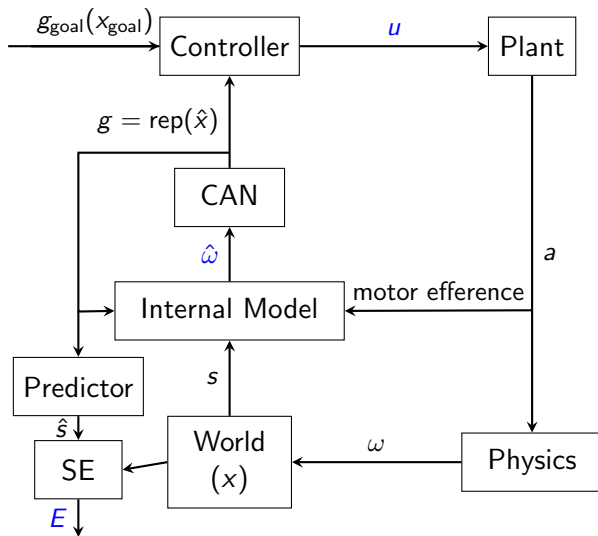
However, this problem does not guarantee any correct alignment with the external world.

The internal model can "dream of" a way to transform motor efference into actions, and the controller can learn to minimize the mismatch between the estimated state and the goal state by this internal model.

A somewhat easy fix is to introduce a predictor which implements $f(g) = \hat{s}$, and force alignment between $\hat{s}$ and s .

To what extent this works is still an open question. I will hopefully get to exploring at least a bit of this tomorrow.

Training the attractor for motor control 2



Training the attractor for motor control 3

The one internal model setup may not work directly. In case it is very hard to train, we can segment the internal model into two parts: one for the motor efference (S) and one for the sensory feedback (S).

The S model is similar to a "critic" here, in that the action that aligns the current internal state and sensory feedback is (monotonically) linked to the error signal aforementioned.

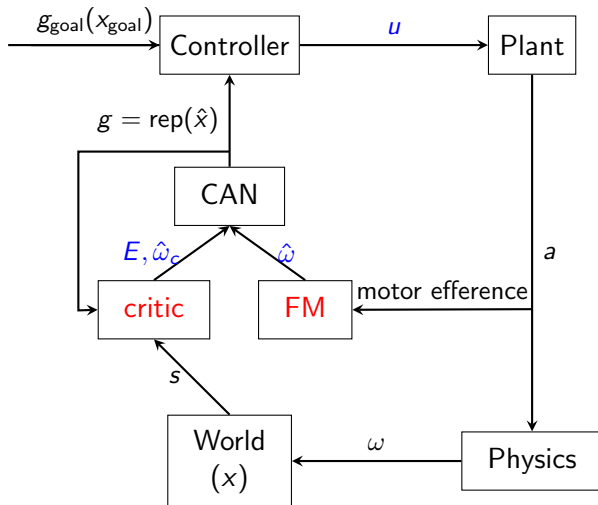
The difference is that, instead of computing only an error signal, the critic also computes a corrective action that updates the internal representation to align with the sensory feedback. Therefore, critic alone can align the internal state with the external world.

Then if you segregate the internal model into the critic and a pure forward model (FM that only takes motor efference), the FM can be trained off the critic's error signal (e.g. to minimize corrective actions or even align actions to corrective actions up to delays).

Also, in egocentric control, the critic is a simple method to drive the CAN to compute and hold the current target transformation (if you always start motor babbling with $g = g_{\text{goal}0}$ and x_0 , the goal state is aligned to the null transformation).

Simply perturbing the sensory feedback from the world results in corrective action that drives the CAN to hold the implied transformation.

Training the attractor for motor control 3



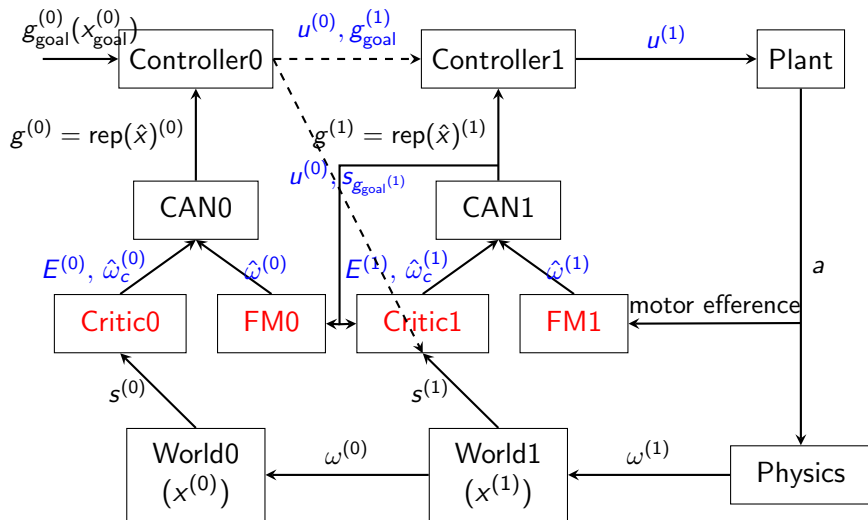
Extension: Hierarchical Attractors for Motor Control

The next page gives a schematic of how the attractor can be extended to a hierarchical system.

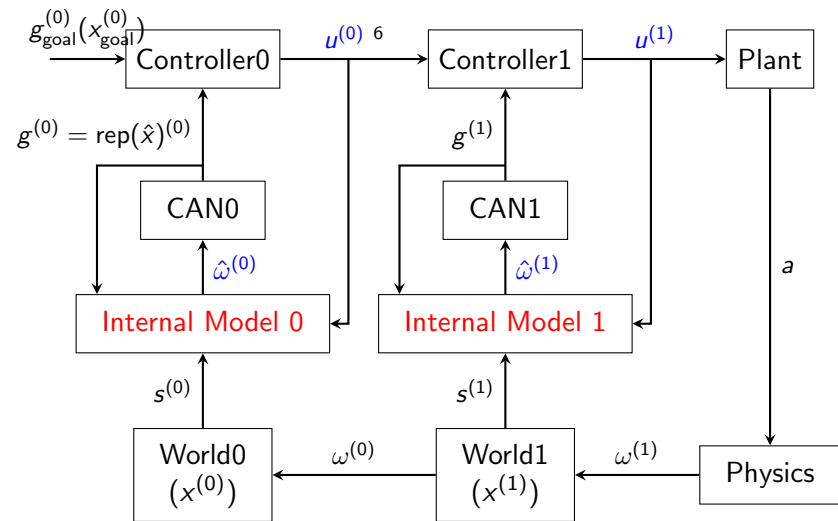
Dash lines from Controller 0 illustrate two ways the controller0 can control the downstream system, corresponding to allocentric and egocentric control.

- allo:** Control by injecting the goal signal directly into the controller1.
- ego:** Control by injecting into the sensory system of the controller 1, the internal model of which then feeds a corrective action to drive CAN to hold the current target transformation.

Extension: Hierarchical Attractors for Motor Control



Extension: Hierarchical Attractors for Motor Control 2



${}^6u^{(0)}$ may convey information about $g_{\text{goal}}^{(1)}$ in the allocentric control case; or equivalently it may be directed to drive the attractor CAN1. But for the latter case another caveat arises from the fact that $g^{(1)}$ must also align with $s^{(1)}$. Therefore, the allocentric may be preferred.

Summary

- ▶ Attractor networks are a powerful tool for representing manifolds in neural circuits.
- ▶ They can be trained with minimal assumptions, and can be used for motor control.
- ▶ The attractor network may be used as an observer for the stimulus manifold in motor control. The full system may be mapped onto a neural circuit for a control problem.
- ▶ My next steps, after training the control circuit, include experimenting with biologically plausible implementations of learning rules. In a longer term, one may consider chained attractor networks for hierarchical control.

Proof of the shift mechanism

let $v(\theta, t) = V(\theta - \omega t)$ be the symmetric travelling profile.

$$V(\theta) + \gamma \frac{dV}{d\theta}(\theta) = [W_{\text{even}} + W_{\text{odd}}] * g(\theta)$$

$$V(\theta) = W_{\text{even}} * g(\theta)$$

$$\gamma \frac{dV}{d\theta}(\theta) = W_{\text{odd}} * g(\theta)$$

$$W * f = \gamma \frac{dW_{\text{even}}}{d\theta} * f$$